

SOVEREIGN QUANT

OFFLINE .

The Architecture of a Self-Owned Quantitative Workstation

A book about the system published at sovereignquant.com.au — multi-agent research, non-bypassable risk, and offline HMAC licensing — written from the live product, not from a generic trading brochure.

Sovereign Quant Australia
v1.3 Core · 2026

Disclaimer

This book describes software. It is not a product disclosure statement, not personal financial advice, and not an offer of a managed fund, brokerage account, or dealing service. Sovereign Quant, as presented on sovereignquant.com.au, is a strategy-research and risk-control workstation that runs on the operator's machine. The public site states plainly that it never holds client money, never places trades for the operator, and does not require a broker account, API keys, or a cloud tenancy to exist as a research desk.

Any backtest figures, Sharpe ratios, Sortino ratios, profit factors, or dashboard screenshots mentioned in this manuscript are illustrations of the reporting surface. They are not forecasts. Past simulated or live results do not guarantee future results. Leveraged markets can produce losses larger than the operator expects, including total loss of trading capital if the operator later connects their own execution stack.

Licence keys described here are software entitlements: Community, Professional, and Institutional tiers that gate capital ceilings, concurrent strategies, and reporting. They are not investment products. HMAC-SHA256 keys are issued after checkout and activated locally. Treat a licence key as you would treat a signing secret.

Where this book distinguishes the public marketing site, the community Streamlit pack served at `/api/download/community`, and the fuller v1.3 agent architecture described in internal project notes, it does so on purpose. A serious book about a real system should not flatten those layers into one mythical platform.

How This Book Was Written

A neighbouring language model was asked to write this book and could not open sovereignquant.com.au. It then invented a nearby company: MetaTrader expert advisors, Myfxbook track records, gold and DAX robots. That is a different object on the public internet. It is not this desk.

This manuscript is written from the live Australian site, the fulfilment terminal that issues HMAC keys, the refund console, the field manual at /guide, the community download that actually returns HTTP 200, and the multi-agent design the product claims. Where something is only a design note and not what the 15-kilobyte community zip contains, the text says so.

The official public surface is <https://sovereignquant.com.au> (www redirects there). A staging preview also exists on Emergent infrastructure; that host is a laboratory, not a customer channel.

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Foreword

Most trading software wants a feed, a cloud login, and a slice of your process. Sovereign Quant was framed the other way around. The homepage does not open with a yield. It opens with a constraint: the workstation runs on your machine. It does not custody funds. It does not place orders on your behalf. It does not ask for broker keys as a condition of existing.

That constraint is easy to market and hard to keep. The moment you add licence checkout, email delivery, analytics, and a concierge widget, you have a server. The honest position — the one this book takes — is that the commercial site is allowed to be online, and the research engine is required to be local. Checkout sees a name and a card. The workstation, once the key is pasted into the sidebar, should not phone home to stay alive.

If you came here for a guru memoir, close the file. If you came here to understand a specific Australian artefact — a licensed offline desk with agents, risk gates, and a fulfilment terminal that prints HMAC keys — continue.

Chapter 1. The Claim on the Door

The live site states its category in one line: Multi-Agent Quantitative Workstation — v1.3 Core. The hero is three words stacked like a warning label: SOVEREIGN. QUANT. OFFLINE.

Under that, the plain sentence that should govern every later chapter: a strategy research and risk-control workstation that runs on your machine. It never holds your money, never places trades for you, and needs no broker account, no API keys, no cloud.

That sentence does two jobs. It tells a buyer what they are not purchasing — not a managed account, not a signal room, not a hosted robot farm. It also tells the builder what they are not allowed to become. The internal project rule is the same as the public rule: offline software licensing is in bounds; a live multi-tenant platform that takes a cut of client transactions is not.

The ticker under the fold

The chrome of the site is deliberate. SYS ACTIVE. HMAC-SHA256 // OFFLINE LICENSING. A pair of coordinates in the corner. A marching line of claims: HMAC-SHA256 offline licensing, non-bypassable risk gates, multi-agent orchestration, kill-switch circuit breaker. This is theatre, but it is theatre attached to real objects: a checkout that requires a licensee name, a fulfilment page that prints a key, a Streamlit sidebar that asks you to activate it.

What the door does not claim

The site does not claim to be ASIC-licensed advice. The login footer says software licensing only, not investment advice. The book follows that line. Architecture can be described with confidence. Edge cannot.

Chapter 2. Why Sovereignty Is an Engineering Problem

Sovereignty here is not a flag or a manifesto adjective. It is a list of refusals.

- Refusal to store the operator's market data on a vendor disk.
- Refusal to make licence validity a daily network round-trip.
- Refusal to let a strategy module bypass the risk module.
- Refusal to take custody so that “ease of use” can be sold.

Each refusal has a cost. Local data means the operator brings files or accepts that optional market fetches happen from their own process. Offline licences mean the vendor cannot instant-revoke a laptop that never calls home — revocation is a commercial and legal problem, not a kill packet. Non-bypassable risk means the convenient backtest button is slower, because every path walks through the same gate.

Cloud is a jurisdiction

If the models live in someone else's region, the desk is only as sovereign as that vendor's terms, that government's warrants, and that outage calendar. The product language — “answers to no server but your own” — is a jurisdictional statement dressed as a tagline.

The commercial website itself is hosted. That is not hypocrisy if the split is clean: the storefront may be rented; the engine may not. The book will keep pointing at that split, because it is where most “offline” products quietly fail.

Chapter 3. The Public Desk

The public desk is a single-page React application with a small set of real routes. The important ones:

- The homepage, with manifesto and pricing as in-page sections.
- `/guide` — the field manual, narrated in six steps, with a two-minute walkthrough video.
- `/login` — licensee access, sign-in and create-account.
- `/payment/success` — fulfilment terminal after Stripe. Without a session it shows SIGNAL LOST, which is the correct failure.
- `/api/download/community` — the free pack, HTTP 200, a zip named sovereign-quant-workstation.zip.
- `/api/payments/checkout` — requires `lookup_key`, `licensee_name`, `origin_url`.
- `/api/auth/me` — returns unauthenticated until a session exists.

Navigation and the deep-link fault

Manifesto and Pricing are homepage anchors. Visiting `/pricing` or `/manifesto` as bare paths currently renders a blank shell. That is a craft defect, not a philosophy. A book can say so without apology. Guide and login are first-class routes and load.

Tone of the desk

Type is display serif for claims, mono for keys and status. Colour moved from an earlier red build to gold and near-black. An earlier concierge widget — Axiom — answered “how do I activate my licence key?” with the `localhost:8501` ritual. Whether that widget is on every production paint, the answer it gave is still the operational truth: run the pack, open the sidebar, paste the key, stay offline.

Australia

The domain is `.com.au`. The company voice is Australian without draping itself in local-finance costume. There is not, on the public pages reviewed for this book, an ABN block or a lengthy privacy policy. That absence should be treated as unfinished corporate hygiene, not as mystique.

Chapter 4. The Super Agent and the Society Beneath It

v1.3 names a society of in-process agents rather than a single script. The public language calls this multi-agent orchestration. The internal map is more specific.

Agent	Job
Orchestrator	Accepts a natural-language goal, routes work, assembles a result with a correlation id.
Data Agent	Loads, caches, and refuses to pretend missing history is a signal.
Strategy Agent	Emits signals from pairs, momentum, or mean-reversion playbooks.
Risk Agent	Applies gates. Can veto. Cannot be skipped by a hot path.
Reporting Agent	Builds branded tearsheets when the licence tier allows.
Licence Agent	Validates the offline key and opens or closes features.

Table 1. The v1.3 agent map as the product describes it.

Messages are typed: GOAL, DATA, SIGNAL, RISK_CHECK, ORDER, REPORT, LICENCE, RESULT, ERROR. The point of types is not fashion. It is so a later log can reconstruct why a veto happened without asking a model to remember.

What “super agent” should mean

It should not mean an unbounded chatbot that wires live orders because a prompt was vivid. It should mean a supervisor that will not let Strategy speak to the outside world until Risk has signed the same correlation id. If a future build grows a conversational overlay, that overlay is a window onto the orchestrator, not a second, softer risk desk.

Entry points

Project notes describe `python scripts/run_agents.py` for the society and `python scripts/run_backtest.py` for the classic single-process path. The community zip the public API actually serves launches Streamlit: `python -m streamlit run app.py` on port 8501. Readers who confuse those two entry points will write the wrong book and ship the wrong support answer.

Chapter 5. Data, Without Surrendering the Books

A quantitative desk is a filing system with opinions. The Data Agent's first virtue is boring: know what you have, cache it where the operator can find it, and do not invent bars.

The community application caches under a local `data_cache` directory and maps a short list of pairs — BTC/USDT, ETH/USDT — onto exchange symbols. Optional fetches against public market endpoints may exist in that pack. That is not the same thing as “your book lives in our cloud.” It is also not the same thing as “zero packets ever leave the machine.” Honesty requires both sentences.

What sovereignty requires of data

- The canonical copy is on disk the operator controls.
- A cache miss is a cache miss, not a silent interpolation dressed as alpha.
- Vendor APIs, if used, are adapters, not landlords.

Pairs research in the fuller architecture is described as cointegration, OLS hedge ratios, and half-life — Engle–Granger territory. Momentum is described with moving averages, ADX, Donchian, and volatility targeting. Mean reversion is a z-score with a regime filter. Those names are research vocabulary. They are not a promise that any one specification is holy.

Chapter 6. The Strategy Engine

The Strategy Playground on the public storyboard is not a supermarket of five hundred indicators. It is three families, because three families force the rest of the system to stay general: a gate that can veto a pairs book must also veto a trend book.

Statistical arbitrage / pairs

Two series, a hedge, a residual, a clock for how fast the residual forgets itself. The work is in the residual's manners, not in the romance of "spread trading." If cointegration is a costume you put on two trending names, Risk should still see a directional book.

Momentum

Trend following is a tax on disbelief. The specification in the notes — moving-average structure, ADX as a participation filter, Donchian as a breakout frame, volatility targeting as the volume knob — is a known grammar. The desk's contribution is not the invention of the moving average. It is that momentum cannot introduce size that Risk did not approve.

Mean reversion

Z-scores without a regime filter are how people donate money to a new trend. The product language includes the filter on purpose. Community tier is described as one concurrent strategy; Professional three; Institutional ten. The cap is not only pricing. It is a confession that attention and correlation are scarce.

Walk-forward and Monte Carlo

Professional copy on the site lists walk-forward analysis and branded tearsheets. Institutional copy lists Monte Carlo and multi-account routing as higher-desk concerns. Treat those as tier gates on research tools, not as proof that a particular Monte Carlo has been run on a particular book this morning.

Chapter 7. Risk Is the Gate, Not a Report

Most platforms print risk after the fact: a red number on a PDF. Sovereign Quant’s public claim is stronger. Risk is a runtime gate. Daily loss, maximum drawdown, portfolio heat, and leverage limits are not suggestions the orchestrator can ignore when a signal looks expensive in the pleasant direction.

Non-bypassable is a software word. It means there is no blessed import path that constructs an order-shaped object without presenting it to RiskAgent. If a future contributor adds a “just this once” flag, they have changed the product into a different product. The book should be used internally as a reason to reject that pull request.

The four ordinary gates

- Daily loss — a clocked budget. Tomorrow is another budget, not a chance to win it back inside the same session by raising size.
- Maximum drawdown — a path budget. Recoveries that require doubling down are not recoveries.
- Heat — how much of the book is simultaneously alive. Correlation is how heat hides.
- Leverage — a multiplier cap that does not care how pretty the z-score is.

A dashboard screenshot circulating with this project showed Community limits in the chrome: a modest max capital and “concurrent strategies: up to 1.” That is the gate made visible. Visibility is not the gate. The gate is the veto.

Chapter 8. The Kill Switch

A kill switch is not a stop-loss. A stop-loss lives on a position. A kill switch lives on the desk. When it is thrown, new risk is refused and live risk is instructed to flatten according to the rules the operator actually wired — which, on the public product, may be “generate no further research actions” rather than “send cancel-replace to a broker,” because the public product does not hold broker keys.

That last clause matters. People who sell kill switches on hosted robots are selling a button that talks to a server that talks to a broker. Sovereign Quant’s switch, if it is faithful to the homepage, is a local circuit breaker. It cannot save an operator who later bolts on their own CCXT adapter and then disables the breaker to “see what happens.”

Project notes mention an execution adapter with sandbox forced by default. That is the correct default. Live keys should feel like breaking glass, not like a toggle.

Chapter 9. Licensing as Cryptography, Not a Paywall Cartoon

The live fulfilment terminal labels the key HMAC-SHA256 // OFFLINE KEY and can mark SIGNATURE VALID. The licensee name is described as being signed into the key so that it appears inside the workstation after activation. Duration on the Professional example is 365 days. Max capital on that tier is one million dollars of notional entitlement inside the software's own ceiling — not a brokerage margin line.

Tier	Price (site)	Max capital	Strategies	Notes from public copy
Community	\$0 forever	\$50,000	1	Evaluate the core. Included in the free download.
Professional	\$499 / year	\$1,000,000	3	Walk-forward, branded tearsheets, signal export.
Institutional	\$1,999 / year	\$50,000,000	10	Monte Carlo, multi-account routing, priority channel.

Table 2. Licence tiers as published on the acquisition section.

HMAC, not a mystical second algorithm

Some draft notes and an older zip README speak of Ed25519. The issued keys and the production chrome say HMAC-SHA256. A book that mixes them will train customers to paste the wrong kind of object into the sidebar. Until the code and the site converge, treat HMAC as the public contract.

What offline activation means

You buy on the site. You receive a key on the fulfilment page and, in the intended design, by email. You paste it under Activate New Licence Key. Verification uses the vendor's secret material at issue time and the public verification logic shipped in the app thereafter. The licensing server is not required to see the machine again for the key to keep working until expiry.

The trade-off is old: a determined operator can abuse a key on several machines; a vendor who never hears from the app cannot push an instant remote revoke to a laptop at the bottom of a drawer. The refund console exists for the commercial half of that problem — refund and revoke on the vendor's ledger — not for reaching into the operator's disk.

Chapter 10. Fulfilment: From Stripe to an Offline Key

The acquisition modal asks for a licensee name before money moves. That is not etiquette. The name is a payload. On a successful Professional checkout the fulfilment terminal has shown a structured record: tier Professional, licensee as typed, duration 365 days, max capital \$1,000,000, a copy-key control, and in some builds a download-workstation control.

If the returning browser has no session, the same route paints SIGNAL LOST and tells the buyer that a charged card still implies an issued licence, recoverable from the receipt. That copy is more adult than a spinner that pretends the world is consistent.

The refund console

An internal page lists orders: licensee, tier, amount, status, and a refund-plus-revoke action on paid rows. Test rows and live rows have coexisted there. That is normal for a desk being born. It is not normal to leave that page reachable without a strong operator secret. Treat /refund as a loaded firearm.

Email

Licence delivery has used transactional email. Addresses on test rows should never be confused with a public support inbox. Customers should recover keys from the licensee portal tied to the email used at checkout, not from screenshots in a group chat.

Chapter 11. Research, Backtests, and How Not to Lie with Numbers

A neighbouring draft quoted dashboard metrics — Sharpe 1.91, Sortino 1.63, profit factor 1.51 — as if they were the firm's audited track record. They are not used that way here. They are the kind of numbers a playground will print when a specification, a window, and a costs model are chosen. Change any of the three and the trophy changes.

Walk-forward exists because a single in-sample peak is a folk tale. Monte Carlo exists because path is not a point estimate. Neither tool forgives a costs model that forgets fees, slippage, and the way liquidity vanishes when the residual finally does what you hoped.

Rules for any figure that leaves the building

- Name the market, the interval, the window, and whether costs were applied.
- Say simulated when it was simulated.
- Never place a Community screenshot next to an Institutional price as if the screenshot were the Institutional book.
- If the public site will not host the figure, the book should not baptise it.

The site's own restraint — no screaming yield on the hero — is the correct editorial standard. Keep it.

Chapter 12. Reporting and the Tearsheet

Institutional culture runs on a one-pager that can survive a cold reading. The product language promises branded tearsheets on paid tiers and locks them on Community. That is both a commercial gate and a pedagogical one: if you have not paid, you are still in the laboratory; do not print a laboratory as letterhead.

A tearsheet that deserves the word should carry the specification hash or name, the licence identity, the gates that were on, and the veto count. A pretty equity curve with the kill switch quietly disabled is not a report. It is an advertisement.

Chapter 13. The Field Manual

The /guide route is titled Zero to Sovereign in Six Steps. It is narrated. It is the onboarding the concierge was already paraphrasing.

The operational sequence, compressed:

- Download the community pack. The proven public command is a GET to `/api/download/community`, which returns a 200 and a 14,976-byte zip.
- On Windows, `run.bat`. On Mac or Linux, `chmod +x run.sh` and `./run.sh`. Python 3 is required. Dependencies install into the local environment.
- Open `http://localhost:8501`.
- Stay on Community, or paste a Professional or Institutional key under Activate New Licence Key.
- Run research inside the gates. Do not confuse a playground button with a broker.
- If you bought a key and the success page died, use the receipt and the licensee email, not a second payment.

The walkthrough video on the guide is two minutes. That is the right length. Anything longer is a course. Anything shorter is a trailer.

Chapter 14. What Ships Today Versus What the Architecture Implies

This chapter is the one a vendor is tempted to delete. It stays.

What a stranger can verify this week: a polished marketing site on sovereignquant.com.au; TLS; a working login surface; a checkout API that validates fields; a community zip that downloads; a Streamlit application with licence chrome, pair pickers, and risk language; a fulfilment aesthetic that has already issued HMAC keys to named licensees; a refund ledger for the operator.

What the Memory Core describes beyond that zip: a Python package tree under `src/sovereign_quant/agents/`, typed messages, `generate_licence.py` as an owner-only tool, paper scripts, a CCXT adapter with sandbox default. That tree is the destination architecture. It is not identical to the 40-kilobyte `app.py` inside the public zip.

A mature company ships the destination or stops advertising it as if it already sat in the customer's downloads folder. Until then, sales copy should say "community workstation" for the zip and "v1.3 architecture" for the package. Customers can live with a roadmap. They cannot live with a file listing that does not match the manifesto.

Chapter 15. Building a Book, a Company, and a Boundary

Sovereign Quant is allowed to be three objects at once, if the labels stay attached:

- A public desk that sells annual offline licences.
- A local research workstation.
- A private architecture that may grow agents, reports, and paper execution.

It is not allowed, by its own rule, to become a hosted dealing room that skims flow. That boundary is easy to recite and expensive to keep when someone asks for “just a small live copy-trader.” The answer is no. The book exists partly so the answer stays no after the founder is tired.

Publishing this book

You do not need a New York house to put a clean manuscript in the world. Practical paths:

- A typeset PDF or print-on-demand edition (Amazon KDP, IngramSpark) with the disclaimer in the front matter, not in a footnote.
- A site edition: one chapter a week under sovereignquant.com.au/book, noindex until the words are finished.
- A licensee edition: the same text bundled in the paid pack, versioned with the software.

Do not send the manuscript to a model that cannot open your domain and then accept its MetaTrader chapter as yours. That episode already happened. It should happen once.

What still wants building on the site

- Deep links for pricing and manifesto.
- The restyled login, if that mock is the intended face.
- A real privacy and terms page on a .com.au commercial site.
- One licence algorithm in all README files.
- A download button on every fulfilment paint, not only some of them.

Chapter 16. The Sovereign Principle

Own the machine that judges the trade. Do not outsource the veto. Do not custody what you do not want to be responsible for. Do not print a backtest as a personality.

The principle is smaller than a vision deck and larger than a feature list. It will still apply if the agents are rewritten, if HMAC is replaced by a better signature, if the only market left on the playground is one unromantic futures contract.

Sovereign Quant, at the domain that actually loads, is an attempt to make that principle purchasable as software without turning it into a fund. This book is an attempt to write that attempt down before the mythology hardens around the wrong product.

Your machine. Your models. Your edge. Your veto.

Glossary

Community pack — The free zip returned by GET /api/download/community.

Fulfilment terminal — The post-checkout page that displays an issued HMAC key.

HMAC-SHA256 — The keyed-hash construction used on the live licence chrome.

Kill switch — Desk-level halt, distinct from a per-position stop.

Licensee name — The string signed into the key and shown after activation.

Non-bypassable gate — A risk check every path must encounter.

Orchestrator — The v1.3 supervisor that routes typed messages among agents.

SIGNAL LOST — The fulfilment state when the browser session cannot be confirmed.

Workstation — Local process, presently launched with Streamlit on port 8501 in the public pack.

Author's Note

This manuscript was commissioned from the operator of sovereignquant.com.au after a draft elsewhere stalled on file limits and on a model that could not retrieve the site. The brief was not “write something that sounds like finance.” The brief was: write the book of this desk.

Where the public product is unfinished, the text says unfinished. Where the architecture is larger than the file customers download, the text says that too. That is the only way a book about sovereignty does not become an advertisement for a cloud.

— Written against the live desk, August 2026.